



Chapter 11: Inheritance and Composition: Part I

Introduction

- Two common ways to relate two classes in a meaningful way are:
 - Composition, or aggregation: (“has-a” relationship)
 - Inheritance (“is-a” relationship)

Composition

Composition (Aggregation)

- In composition, one or more member(s) of a class are objects of another class type
- Composition (aggregation): “has-a” relation
 - Composition is a strong relationship (e.g., University and Departments)
 - Aggregation is a weak relationship (Department and Instructors)

Composition (Aggregation)

- Arguments to the constructor of a member-object are specified in the heading part of the definition of the constructor
- Member-objects of a class are constructed in the order they are declared
 - Not in the order listed in the constructor's member initialization list

Composition (Aggregation)

```
#include<iostream>
using namespace std;
class Door
{
public:
    Door(){state=0;}
    Door(int s) {state=s;}
    void setState(int s) {state=s;}
    int getState() {return state;}
    void print()
    {
        if (state)
            cout<<"Open"<<endl;
        else
            cout<<"Close"<<endl;
    }
private:
    int state;
};
```

```
class Room
{
public:
    Room(){l=w=0;}
    Room(int ll, int ww, int state): door(state)
    { l=ll; w=ww;}
    void setLW(int ll, int ww)
    {
        l=ll; w=ww;
    }
    int getL() {return l;}
    int getW() {return w;}
    void openDoor() {door.setState(1);}
    void closeDoor() {door.setState(0);}
    void print()
    {
        cout<<"Length: "<<l<<endl;
        cout<<"Width: "<<w<<endl;
        door.print();
    }
private:
    int l,w;
    Door door;
};
```

```
int main()
{
    Room r (5,3,0);
    r.print();

    r.openDoor();
    r.setLW(10,5);
    r.print();

    //r.door.setState(1);// ERROR:

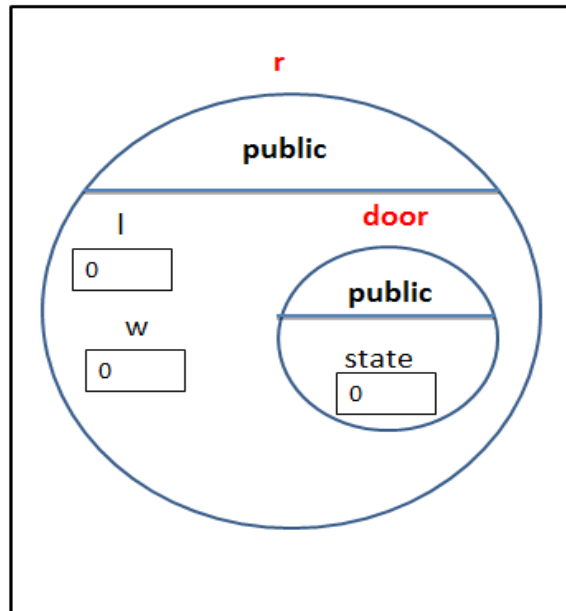
    return 0;
}
```

OUTPUT

```
Length: 5
Width: 3
Close
Length: 10
Width: 5
Open
```

Order of Construction & Destruction

- They are constructed before the containing class objects are constructed
- The order of destruction is the opposite order of construction



Composition & Default Constructor

- If you do not initialize a member object in the member initializer list of the constructor (or if the constructor is a compiler-generated constructor), the compiler will automatically invoke the member object's default constructor
- If the member object's default constructor is missed and other constructors are available, an error will result

Graphical Representation of Composition & Aggregation

Aggregation



Composition



Summary

- In composition (aggregation):
 - Class member is an object of another class
 - Call to constructor of member objects is specified in heading of the definition of class's constructor